TOA- Assignment 2

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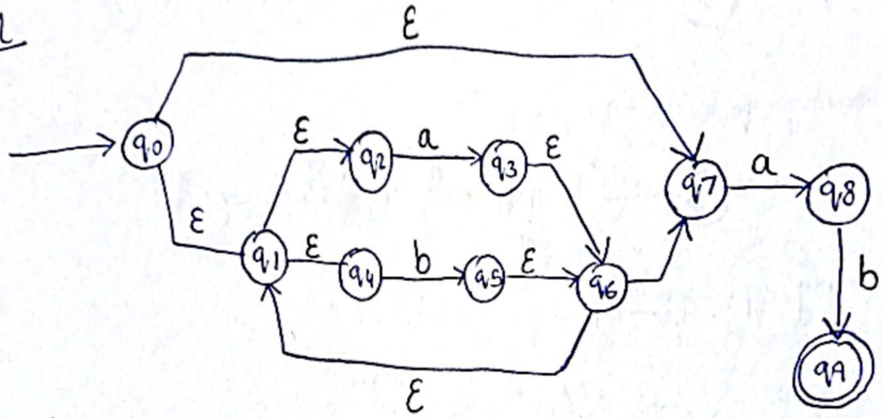
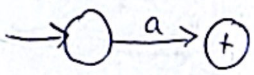
Section A :-

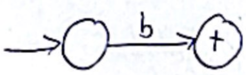
Q1a) Regular Expression :- $(a+b)^*ab$

sol

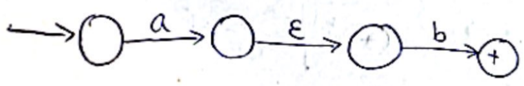


working of sub diagrams

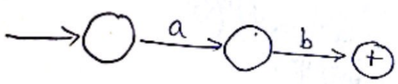




so ab (concatenate)

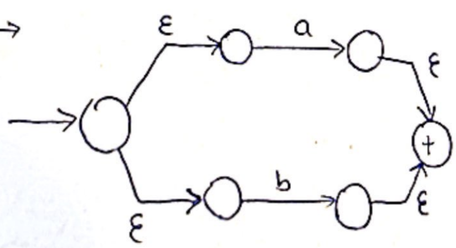


simplify

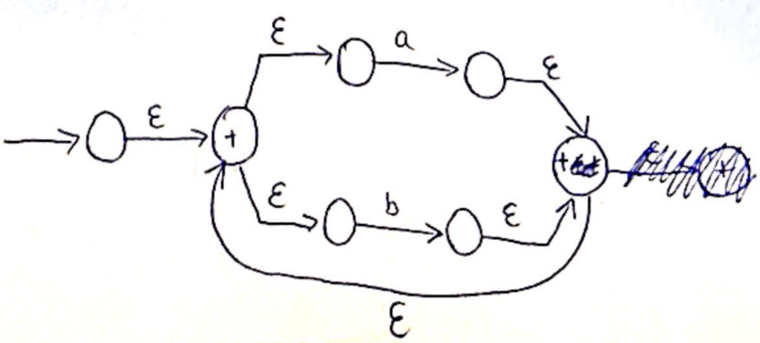


for $(a+b)^*$

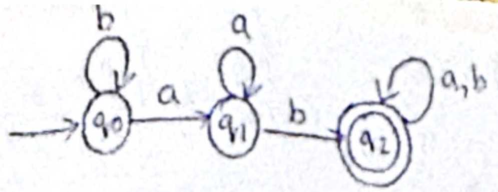
a+b →



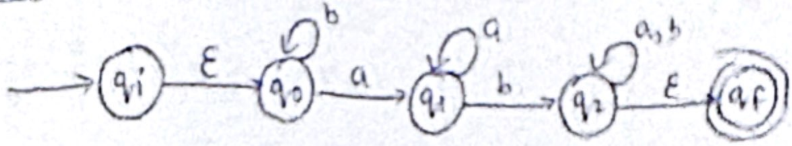
now $(a+b)^*$



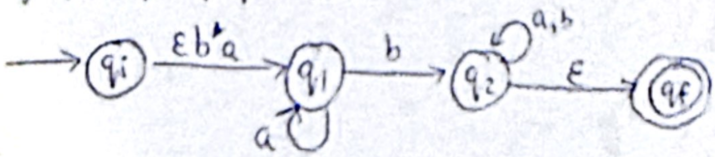
(Q1b)



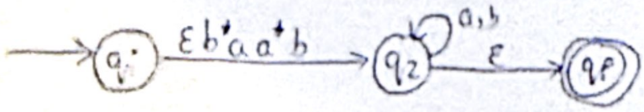
working



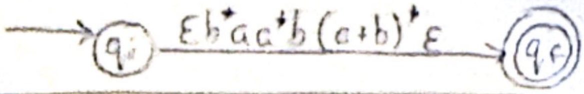
Removing q_0: $q_i - q_0 q_1 = \varepsilon b^* a$



Removing q_1: $q_i - q_1 - q_2 = \varepsilon b^* a \cdot a^* b$

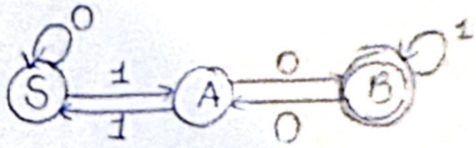


Removing q_2: $q_i - q_2 - q_f = (\varepsilon b^* a a^* b) \cdot (a+b)^* \varepsilon$

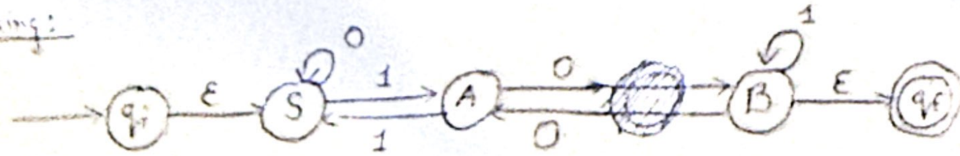


$\boxed{\therefore \text{ Regular Expression : } b^*aa^*b(a+b)^*}$ Ans

(Q2a)



working:



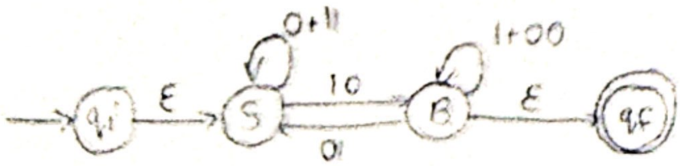
Removing A:

SAB : 10

BAS : 01

SAS : 11

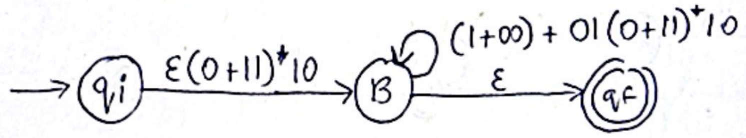
BAB : 00



Removing S:

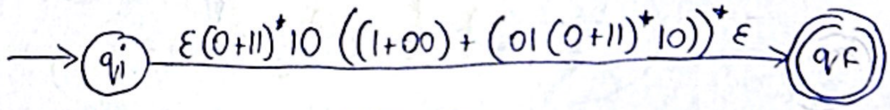
$q_i - S - B = \varepsilon(0+11)^*10$

$B - S - B = (1+00) + 01(0+11)^*10$



Removing B:

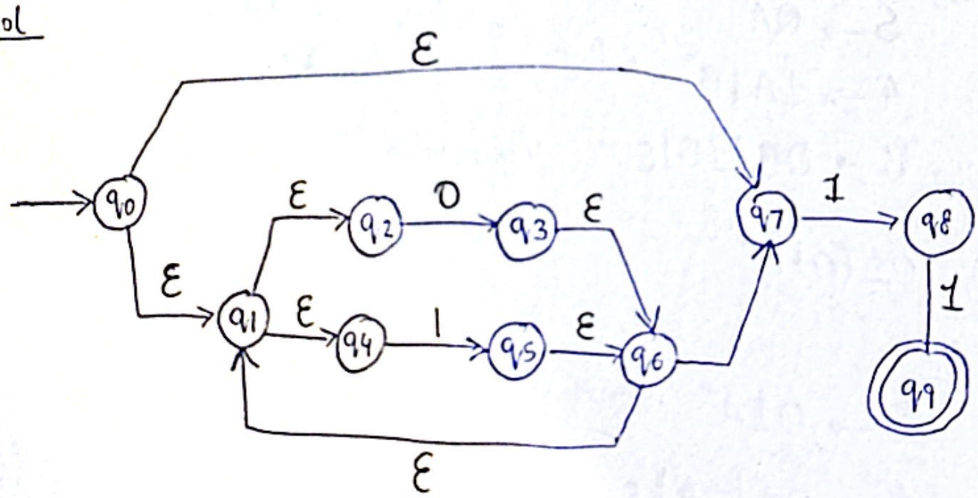
$q_i - B - q_f = \varepsilon(0+11)^*10((1+00)+(01(0+11)^*10))^*\varepsilon$



$\boxed{\therefore \text{Regular Expression}: (0+11)^*10((1+00)+(01(0+11)^*10))^*}$ Ans

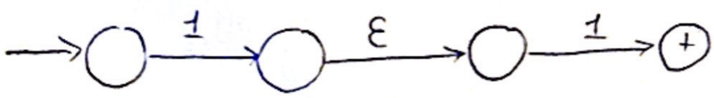
Q2b) Regular Expression :- $(0+1)^*11$

Sol

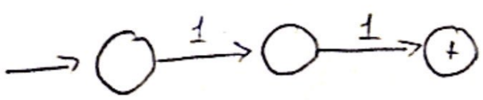


working of sub diagram

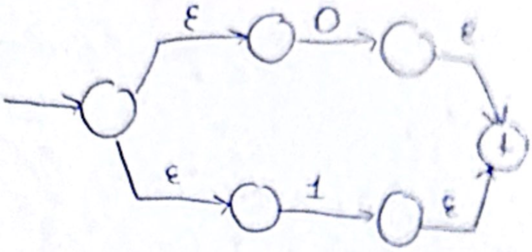
so 11 concatenation



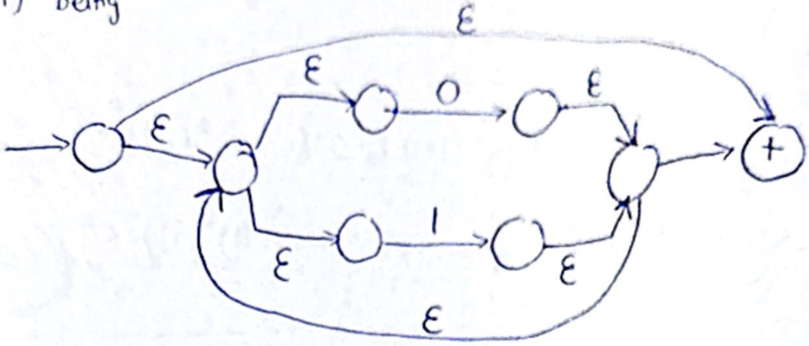
simplify



for (0+1)



∴ $(0+1)^*$ being



Section B:

Q1 i) $01^* [0+1]^*$

sol

$S \rightarrow 0A$ // will begin with 0 always

$A \rightarrow 1A \mid B$ // will generate 1^* or jump to B

$B \rightarrow 0B \mid 1B \mid \varepsilon$ // Generates $(0+1)^*$ ends in ε

ii) $01(0+1)^*$

sol

$S \rightarrow 01A$

$A \rightarrow 0A \mid 1A \mid \varepsilon$

iii) Letters followed by digits

$L = \{xy \mid x \in \{a \ldots z, A \ldots Z\} \; y \in \{0-9\}\}$

my assumption string is of two length, and small + capital letter

sol

$S \rightarrow aQ_0 \mid bQ_0 \mid \ldots \mid zQ_0 \mid AQ_0 \mid BQ_0 \mid \ldots \mid ZQ_0$

$Q_0 \rightarrow 0 \mid 1 \mid 2 \mid 3 \mid 4 \mid 5 \mid 6 \mid 7 \mid 8 \mid 9$

iv) Start and end with a.

Regular expression: $a(a+b)^*a$

assuming Alphabet $\{a,b\}$

Sol

$S \rightarrow aA \mid a$ // always start with a, will accept single a

$A \rightarrow aA \mid bA \mid a$ // $(a+b)^*$ displays multiple time will always terminate at a

accepted string = (a, aaa, aababa ....)

V) Generate all even integer upto 998

eg strings (0, 2, 4, ... , 10, 12, 14, ... 102, 104, ... 312, 396, ... 998)

sol

$S \rightarrow 0 \mid X \mid NE \mid NDE$

$X \rightarrow 2 \mid 4 \mid 6 \mid 8$

$N \rightarrow 1 \mid 2 \mid 3 \mid 4 \mid 5 \mid 6 \mid 7 \mid 8 \mid 9$

$E \rightarrow 0 \mid 2 \mid 4 \mid 6 \mid 8$

$D \rightarrow 0 \mid 1 \mid 2 \mid 3 \mid 4 \mid 5 \mid 6 \mid 7 \mid 8 \mid 9$

where

X is single digit number being printed

NE is double digit number (10, 12, 42, 46, 66, 78, 98)

NDE is triple digit number (106, 108, 612, 746, 908, 998)

vi) Alternate Letters

~~Assumpti~~ Assuming alphabet $\{a,b\}$

so Regular Expression : $(ab)^* + (ba)^*$

eg strings (ab, ababab, ba, baba, ...)

Sol

$S \rightarrow A \mid B$

$A \rightarrow abA \mid \varepsilon$ // will give $(ab)^*$

$B \rightarrow baB \mid \varepsilon$ // will give $(ba)^*$

vii) a^{2n}

$L = \{a^{2n} \mid n \geq 0\}$

so Regular expression = $(aa)^*$

Sol

$$S \rightarrow aaS \mid \varepsilon$$

S
|
n=1 aaS
|
n=2 aaS
|
n=3 aaS

S
|
ε

viii) $[ab]^n$

$L = \{(ab)^n \mid n \geq 0\}$

so Regular expression = $(ab)^*$

Sol

$$S \rightarrow abS \mid \varepsilon$$

ix) Generate {012, 120, 201}

Sol 1

$$S \rightarrow 012 \mid 120 \mid 201$$

Sol 2

$S \rightarrow 0A \mid 1B \mid 2C$

$A \rightarrow 1D$

$D \rightarrow 2$

$B \rightarrow 2E$

$E \rightarrow 0$

$C \rightarrow 0F$

$F \rightarrow 1$

x) Grammar for the set of all strings over {0, 1} consisting of equal number of 0's and 1's

$$S \rightarrow 0S1 \mid 1S0 \mid 10S \mid 01S \mid \varepsilon$$

Q2a) $S \rightarrow aA \mid bB$

$A \rightarrow aA \mid bS \mid \varepsilon$

$B \rightarrow bB \mid aS$

working:

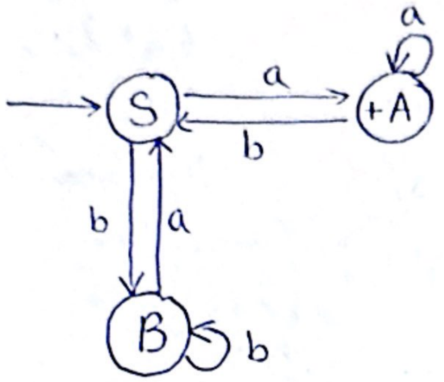
Initial State : S

Non-terminal : {S, A, B}

Terminal: {a, b)

Final : A as ε terminates

FA:



δ	a	b
S	A	B
A	A	S
B	S	B

b) $S \rightarrow 0S \mid 1A$

$A \rightarrow 0A \mid 1$

~~B~~

Sol

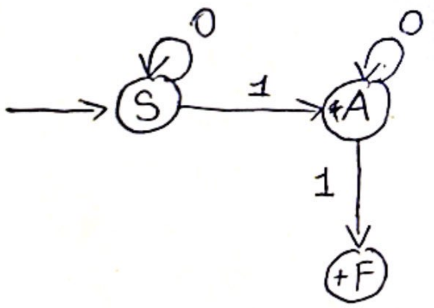
working:

Initial State : S

Final State : ~~A~~ as 1 terminate

Non-terminal : {A, S, F}

Alphabet/Terminal : {0, 1)



δ	0	1
S	S	A
A	A	F
F		

c) $S \rightarrow 0A \mid 1B$

$A \rightarrow 1S \mid 0A$

$B \rightarrow 0S \mid 1B$

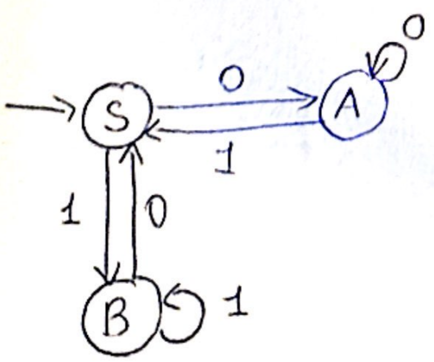
working:

Initial State : S

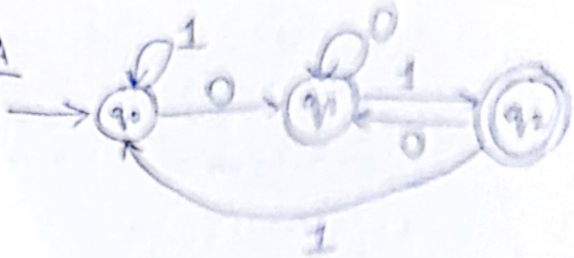
Non-terminal State : {S, A, B}

Alphabet : {0, 1}

Final state :- none as no terminal or epsilon end, infinite loop



Q3a) DFA



Start State : Q_0
Final State : Q_2
Alphabet = {0, 1}
Non-terminal = {Q_0, Q_1, Q_2

Right-Linear Grammar [RLG]

$Q_0 \rightarrow 0Q_1 \mid 1Q_0$

$Q_1 \rightarrow 0Q_1 \mid 1Q_2$

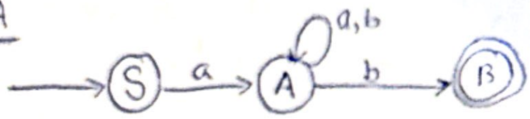
$Q_2 \rightarrow 0Q_1 \mid 1Q_0 \mid \varepsilon$

CFG

$S \rightarrow X01$

$X \rightarrow 0X \mid 1X \mid \varepsilon$

b) NFA



Start State : S
Final State : B
Alphabet = {a, b}
Non-terminal = {S, A, B}

Right-Linear Grammar [RLG]

$S \rightarrow aA$

$A \rightarrow aA \mid bA \mid bB$

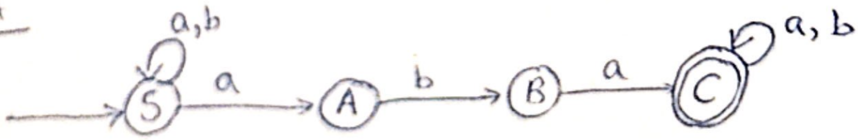
$B \rightarrow \varepsilon$

CFG

$S \rightarrow aMb$

$M \rightarrow aM \mid bM \mid \varepsilon$

c) NFA



Start State : S
Final State : C
Alphabet : {a, b}
Non-Terminal = {S, A, B, C}

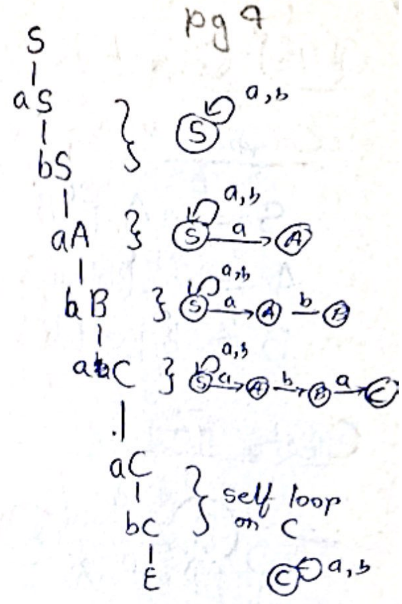
Right Linear Grammar (RLG)

$S \rightarrow aS \mid bS \mid aA$

$A \rightarrow bB$

$B \rightarrow aC$

$C \rightarrow aC \mid bC \mid \varepsilon$



CFG

$S \rightarrow XabaX$

$X \rightarrow aX \mid bX \mid \varepsilon$

Q4a) Converting LLG to RLG

Original LLG

$S \rightarrow Aa \mid Ab$

$A \rightarrow Ba \mid Bb \mid a$

$B \rightarrow b$

Find string produced by LLg

$S \rightarrow Aa \mid Ab$

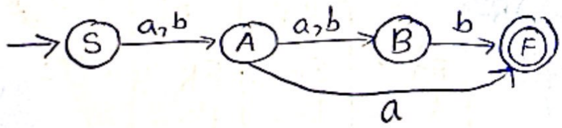
$A \rightarrow ba \mid bb \mid a$

so

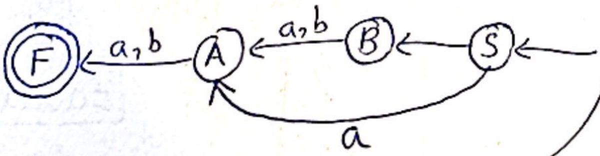
$S \rightarrow baa \mid bba \mid aa \mid bab \mid bbb \mid ab$

$L = \{aa, baa, bba, ab, bab, bbb\}$ generated

Create FA



Reverse FA



To check if RLG was correcty converted let's confirm using parse tree

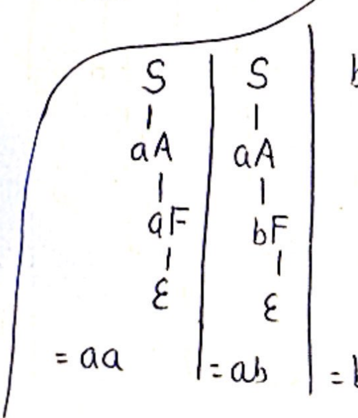
RLG:

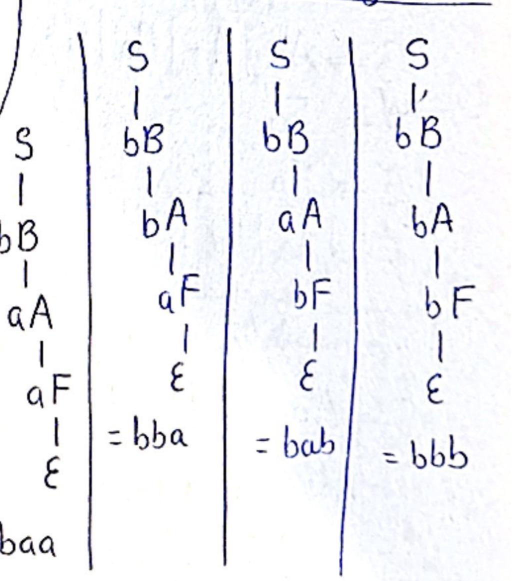
$S \rightarrow aA \mid bB$

$A \rightarrow aF \mid bF$

$B \rightarrow aA \mid bA$

$F \rightarrow \varepsilon$





∴ The conversion was successful as strings made by LLG matches with strings made by RLG. You can see parse tree

Q4b) Convert the following RLG → LLG

Original RLG

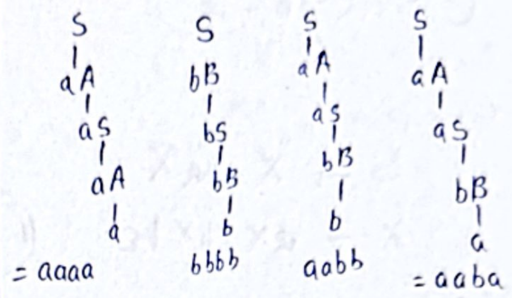
$S \rightarrow aA \mid bB$

$A \rightarrow a \mid b \mid aS$

$B \rightarrow b \mid a \mid bS$

Find some strings made by RLG to confirm with LLG string

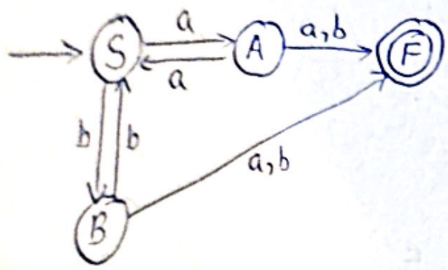
{aa, ab, ba, bb, aaaa, bbbb, aabb, aaba



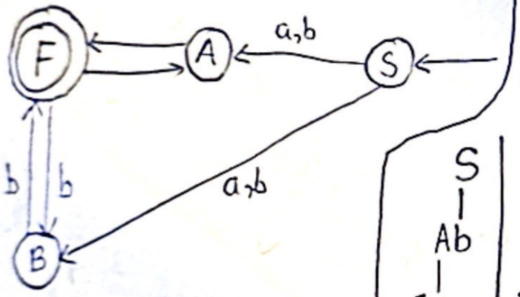
∴ Test case string

aa, ab, ba, bb, aaaa, bbbb, aabb, aaba

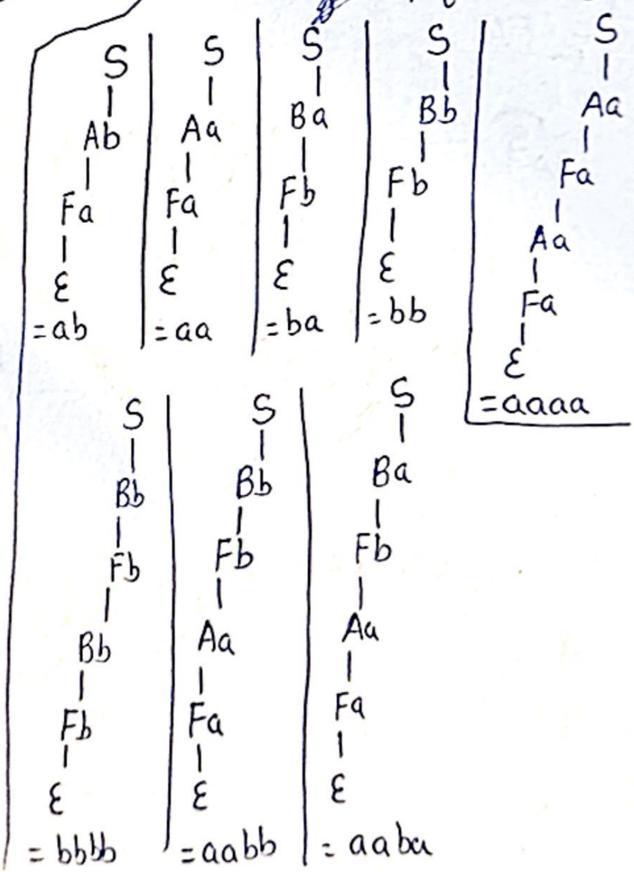
Create FA



Reverse FA



To check if LLG was correctly converted, let's confirm using parse tree. (~~using simplified LLG~~)



LLG:

$S \rightarrow Aa \mid Ab \mid Ba \mid Bb$

$A \rightarrow Fa$

$B \rightarrow Fb$

$F \rightarrow Aa \mid Bb \mid \varepsilon$

if we simplify it then

$S \rightarrow Aa \mid Ab \mid Ba \mid Bb$

$A \rightarrow a \mid Aaa \mid Bba$

$B \rightarrow b \mid Aab \mid Bbb$

∴ The conversion was successful as strings made by RLG matches with string made by LLG. You can see parse tree.

Q5) For 1

$S \rightarrow aSb \mid ab$ string = aabb

a) substitution

$S \rightarrow aSb$
$a(ab)b$
$aabb$

String achieved

b) Left most derivation

$S \rightarrow aSb$
$a(ab)b$
$aabb$

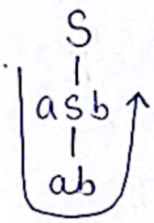
String achieved

Right most derivation

$S \rightarrow aSb$
$a(ab)b$
$aabb$

String achieved

c) Parse tree



= aabb

String achieved

Note = All three derivation coincide cause at each stage there is just one non-terminal

Q5) For 2

$S \rightarrow aS \mid Sb \mid ab$ string = aaabb

a) substitution

$S \rightarrow aS$
$a(aS)$
$aa(Sb)$
$aa(ab)b$
$aaabb$

String achieved

b) Left most derivation

$S \rightarrow Sb$
$(aS)b$
$a(aS)b$
$aa(ab)b$
$aaabb$

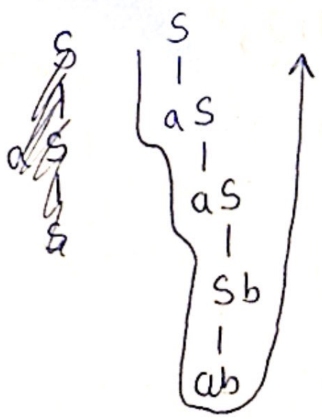
String achieved

Right most derivation

$S \rightarrow aS$
$a(aS)$
$aa(Sb)$
$aa(ab)b$
$aaabb$

String achieved

c) Parse tree



= aaabb

String achieved

Q5) For 3

$S \rightarrow 0S1 \mid 01$ string 000111

a) Substitution

$S \rightarrow 0S1$

$0(0S1)1$

$00(01)11$

000111

string achieved

b) Left most derivation

$S \rightarrow 0S1$

$0(0S1)1$

$00(01)11$

000111

string achieved

Right most derivation

$S \rightarrow 0S1$

$0(0S1)1$

$00(01)11$

000111

string achieved

c) Parse Tree

S
|
0S1
|
0S1
|
01

= 000111

string achieved

Q6i) CGF for balanced parenthesis in this alphabet set {(,)}

sol

CGF ⇒ $S \rightarrow (S) \mid SS \mid \varepsilon$ eg string : (), (()()) ...

iia) CGF of even palindrome 001 100

sol

CGF ⇒ $S \rightarrow 0S0 \mid 1S1 \mid \varepsilon$

eg string : 001100, 010010, 011110, ...

iib) CGF of odd palindrome 00100

sol

CGF ⇒ $S \rightarrow 0S0 \mid 1S1 \mid 0 \mid 1$

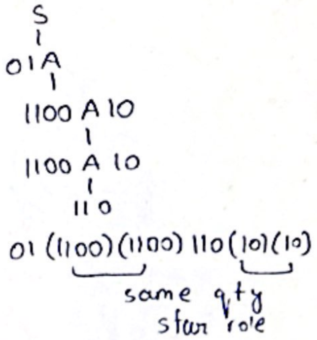
eg string : 01010, 01110, 00100, ...

) $\{01(1100)^n 110(10)^n \mid n \geq 0\}$

sol

$S \to 01A$

$A \to 1100A10 \mid 110$



) $\{0^n 1^m \mid n \neq m\}$

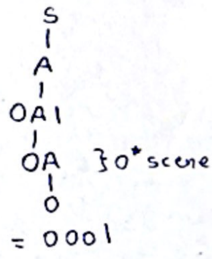
sol

so $n > m$ and $n < m$ means 0 more than 1 and 1 more than 0.

$S \to A \mid B$

$A \to 0A1 \mid 0A \mid 0$

$B \to 0B1 \mid B1 \mid 1$



g string ; 0, 1, 001, 011, 00*01, 01*11, ...

) $\{[0^n 1^n 2^m 3^m \mid n>=1, m>=1] \text{ or } [0^n 1^m 2^m 3^n] \mid n>=1, m>=1]\}$

sol Language 1 Language 2

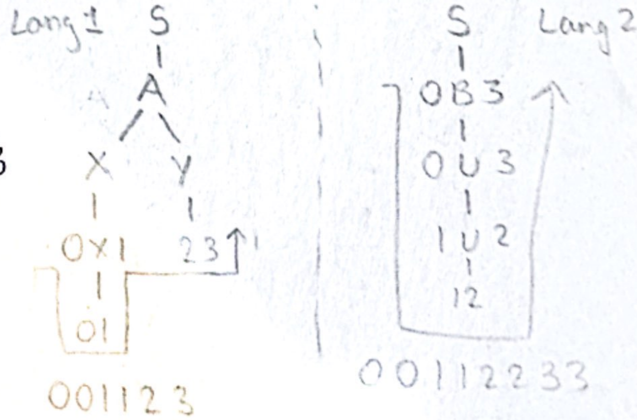
union of it in $S \to L_1 \mid L_2$

for Language 1

$A \to XY$

$X \to 0X1 \mid 01$ // generates $0^n 1^n$ $n>=1$

$Y \to 2Y3 \mid 23$ // generates $2^m 3^m$ $m>=1$



for Language 2

$B \to 0B3 \mid 0U3$

$U \to 1U2 \mid 12$

∴ CFG

$S \to A \mid B$ // this will call language 1 grammar or language 2 grammar

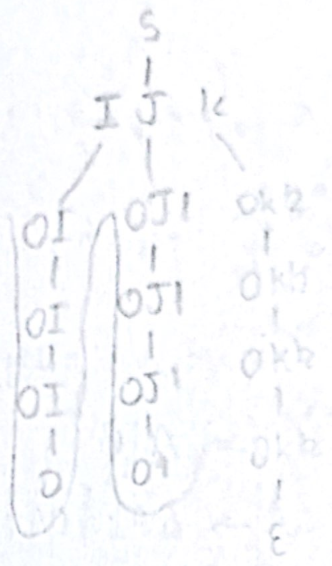
vi) $\{[0^i 1^j 2^k \mid k <= i, k <= j]\}$

$S_1 \to PQ$

$P \to 0P \mid \epsilon$

$Q \to 0Q1 \mid R$

$R \to 1R2 \mid \epsilon$



0 greater

CGF for $L_1 (i > j > k)$

$S \to 0S1 \mid T$

$T \to 0T2 \mid U$

$U \to 0U1 \mid 01$

Part vi, vii, x are not context free so they are not solvable. Therefore skipping them

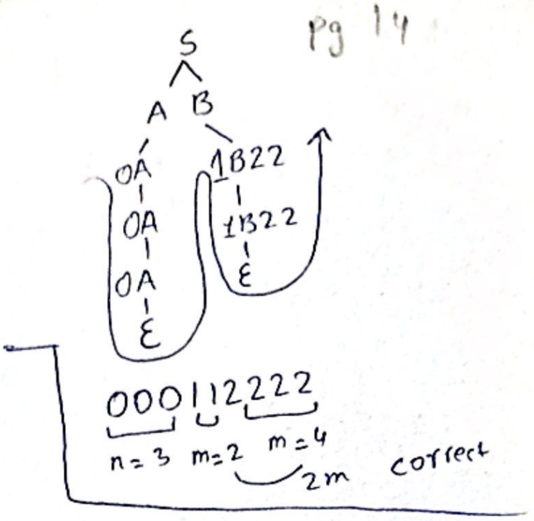
(viii) $\{0^n 1^m 2^{2m} \mid n, m \geq 0\}$

sol

CGF ⇒ $S \to AB$

$A \to 0A \mid \varepsilon$

$B \to 1B22 \mid \varepsilon$

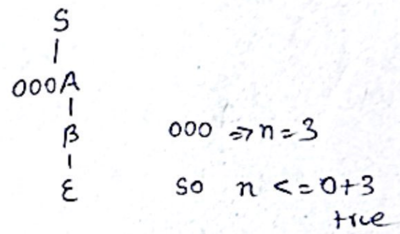


x) $\{0^n 1^m \mid n \leq m+3\}$

$S \to 000A \mid 00A \mid 0A \mid A$

~~$A \to 0A1 \mid B$~~ $A \to 0A1 \mid B$

~~$B \to B1B \mid \varepsilon$~~ $B \to 1B \mid \varepsilon$



7e) Input alphabet = $\{[,], \{, \}, (,)\}$

CFG ⇒ $S \to (S) \mid [S] \mid \{S\} \mid SS \mid \varepsilon$

Section C

CGF ⇒ $S \to aSa \mid bSb \mid a \mid b$

$A \to B \mid a$

$B \to C \mid b$

1) Removing useless production from CFG

non-terminal states A, B, C are not reachable from S so removing them

$\therefore \quad S \to aSa \mid bSb \mid a \mid b$

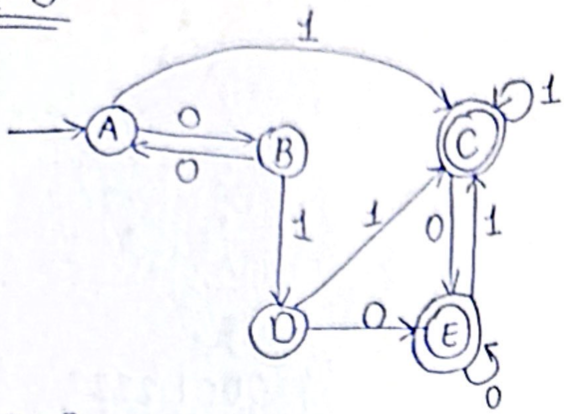
) Check for ambiguity in the given CFG for the string 'abba'

The CFG generates odd length palindrome, while string "abba" is of even length. Therefore it ain't in the language.

As this string can't be generated, the question of ambiguity

Section D

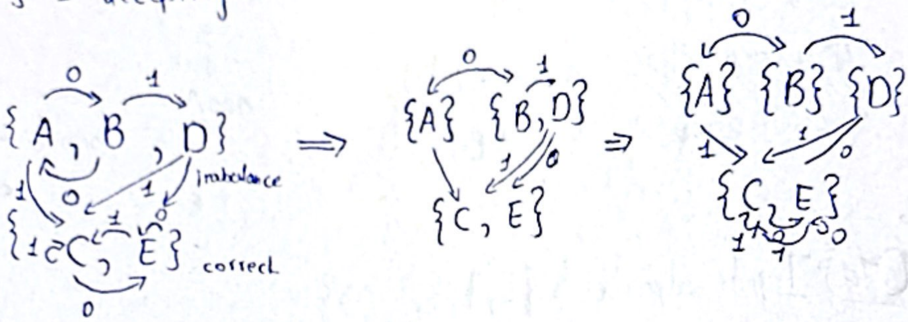
Q1)



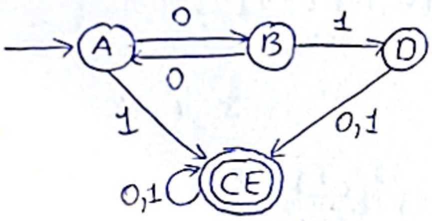
$\{A, B, D\}$ = Non-accepting

$\{C, E\}$ = accepting

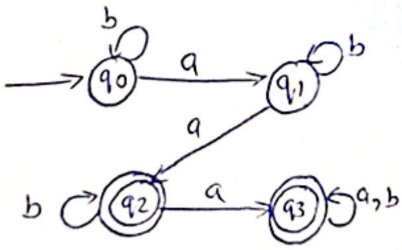
Sol



Minimised DFA

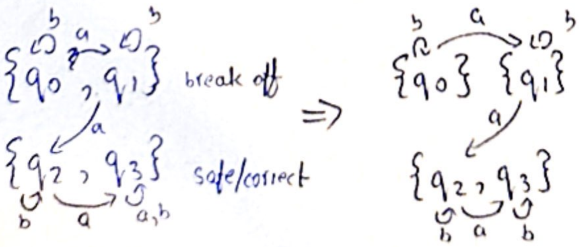


Q2) DFA



Accepting = $\{q_2, q_3\}$

Non-accepting = $\{q_0, q_1\}$



minimized successful

Minimised DFA